

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2015

Subject: 353152 Electronic Circuits I

Section: 5-6

Date: 12 October 2015

Time: 15.00-17.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Text books are **NOT** allowed.
 4. Calculator is allowed.
 5. Electronic communication device are **NOT** allowed in the examination room.
 6. The examination has 10 pages (including this page), 2 parts (35 questions) and a total score of 85 points.
 7. Write all your answers on this examination sheet.
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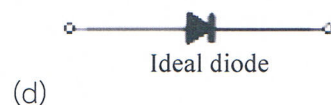
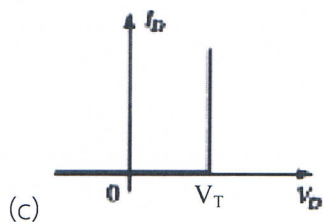
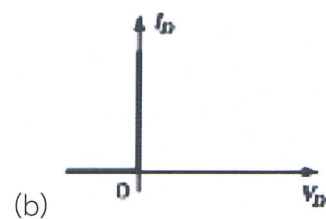
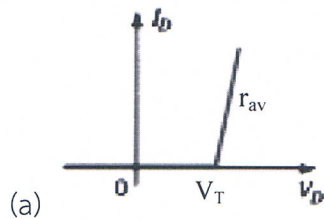
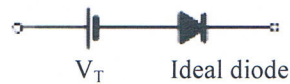
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Part 1 (60 points) Multiple choices

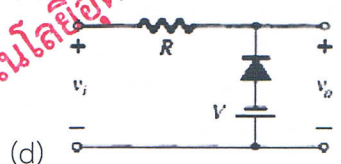
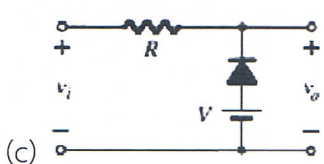
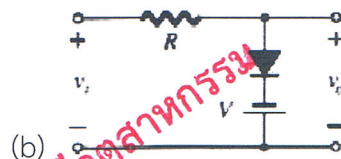
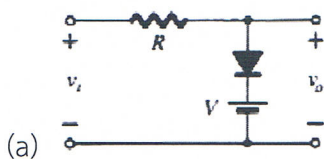
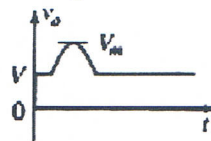
Instruction: Mark the correct answer for the following questions in the provided answer sheet.

1. In the atomic lattice the _____ and _____ form the nucleus.
 (a) electrons, neutrons (b) electrons, protons
 (c) neutrons, protons (d) nucleon, positron
2. Valence shell contains up to _____ electrons.
 (a) 1 (b) 2 (c) 4 (d) 8
3. In p-type material the _____ is called the majority carrier.
 (a) electron (b) hole (c) proton (d) neutron
4. In what state is a germanium diode if the voltage drop across it is about 0.3 V?
 (a) no bias (b) forward bias (c) reverse bias (d) zener region
5. The source voltage must be _____ the voltage drop across the diode to conduct the diode.
 (a) smaller than (b) larger than (c) the same as (d) zero
6. Which type of material has the smallest energy gap between conduction band and valence band?
 (a) conductor (b) semiconductor (c) superconductor (d) insulator
7. Which quadrant of diode characteristic graph is in reverse bias operation?
 (a) 1 (b) 2 (c) 3 (d) 4
8. How much is the voltage drop over germanium diode when it is in forward bias mode?
 (a) 0.2 V (b) 0.3 V (c) 0.6 V (d) 0.7 V
9. How much is the resistance of a diode when it is in forward bias mode for the ideal case?
 (a) 0 Ω (b) 0.3 Ω (c) 0.7 Ω (d) infinite
10. Calculate DC resistance R_{DC} of a diode having $I_D = 30$ mA and $V_D = 1.2$ V.
 (a) 25 Ω (b) 40 Ω (c) 0.04 Ω (d) 0.025 Ω

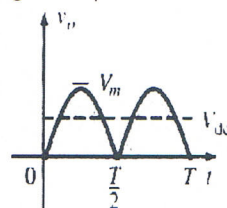
11. Which one is corresponding to the following figure?



12. Which circuit can generate output voltage as below?



13. Which of the following is the average output value ?



(a) $0.318V_m$

(b) $0.5V_m$

(c) $0.636V_m$

(d) $1.414V_m$

14. The intersection of the load line with the characteristic curve determines the _____ of the system.

(a) characteristic curve

(b) forward bias

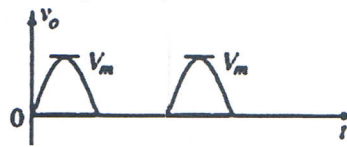
(c) load-line analysis

(d) point of operation

15. Which one the of following is true?

- (a) Diode is open circuit when operating in forward bias mode.
- (b) Forward bias means the voltage on p junction is lower than n junction.
- (c) For ideal diode, there is no voltage drop for forward bias mode.
- (d) No correct answer.

16. Which of the following is the average output value ?

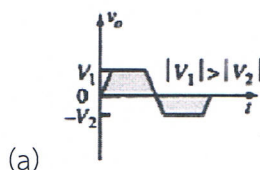
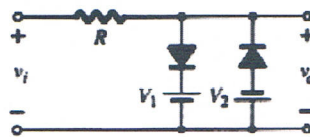


- (a) $0.318V_m$
- (b) $0.5V_m$
- (c) $0.636V_m$
- (d) $1.414V_m$

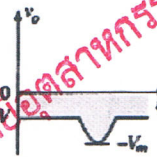
17. The load line is defined by the _____ and a characteristic curve is defined by the _____.

- (a) quiescent point, device
- (b) device, network dc network
- (c) network, device
- (d) None of the above

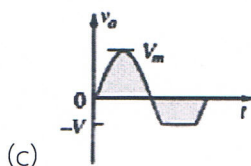
18. What is the output voltage of circuit below if $v_i = V_m \sin(\omega t)$ [volts] ?



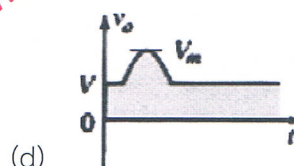
(a)



(b)



(c)



(d)

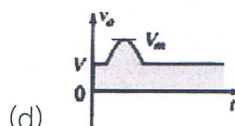
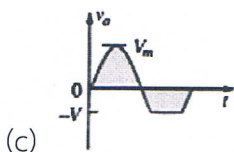
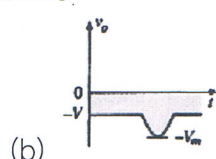
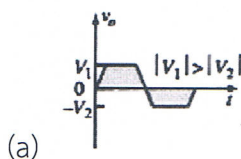
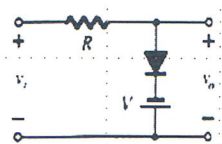
19. Which of the following is true?

- (a) EB Junction is forward biased when transistor is in active mode.
- (b) EB Junction is forward biased when transistor is in cutoff mode.
- (c) CB Junction is reverse biased when transistor is in saturation mode.
- (d) CB Junction is forward biased when transistor is in cutoff mode.

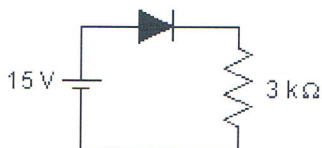
20. Which of following transistor operating mode represents the open circuit?

- (a) Active
- (b) Cut-off
- (c) Saturation
- (d) None

21. What is the output voltage of circuit below if $v_i = V_m \sin(\omega t)$ [volts]?



22. Determine the current level if $E = 15 \text{ V}$ and $R = 3 \text{ k}\Omega$, if the diode is assumed to be an ideal diode.



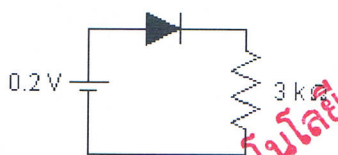
(a) 0 A

(b) 4.76 mA

(c) 5 mA

(d) 5 A

23. Determine the voltage across the resistor, if the diode is assumed to be an ideal diode.



(a) 0 V

(b) 0.09 V

(c) 0.2 V

(d) 0.44 V

24. Zener diodes are used in regulator networks to ____.

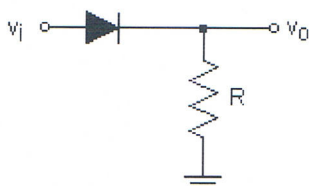
(a) generate voltage

(b) consume power

(c) maintain a fixed voltage across the load resistor

(d) protect the load

25. What best describes the circuit?



(a) Clamper

(b) Bridge rectifier

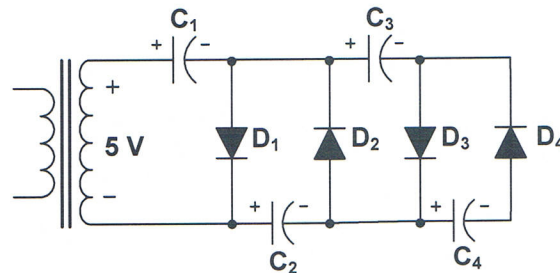
(c) Clipper

(d) Full-wave rectifier

26. Which of the following equipment can check the condition of a transistor?

- (a) Current tracer (b) Digital display meter (DDM)
(c) Ohmmeter (VOM) (d) All of the above

27. What is the voltage measured from the negative terminal of C_4 to the negative terminal of the transformer?

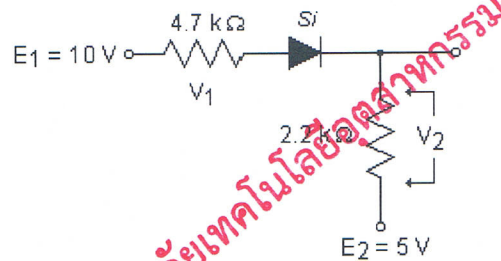


- (a) -20 V (b) -10 V (c) 10 V (d) 20 V

28. Which of the following currents is nearly equal to each other?

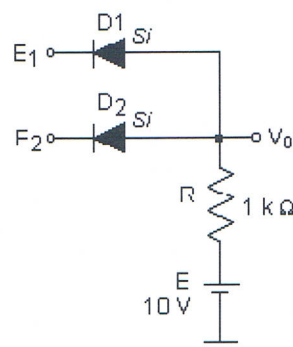
- (a) I_B and I_C (b) I_E and I_C (c) I_B and I_E (d) I_B , I_C , and I_E

29. Determine V_2 .



- (a) 3.201 V (b) 0 V (c) 4.3 V (d) 1.371 V

30. Determine V_o if $E_1 = E_2 = 10$ V.

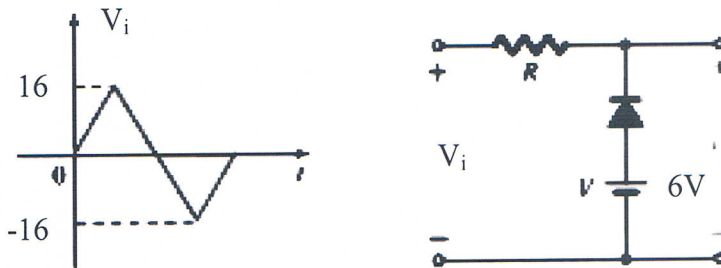


- (a) 9.3 V (b) 10 V (c) -10 V (d) 0 V

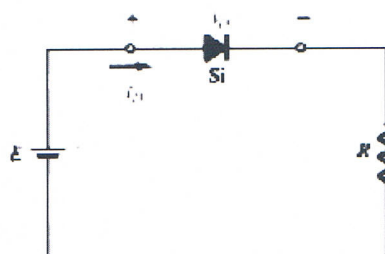
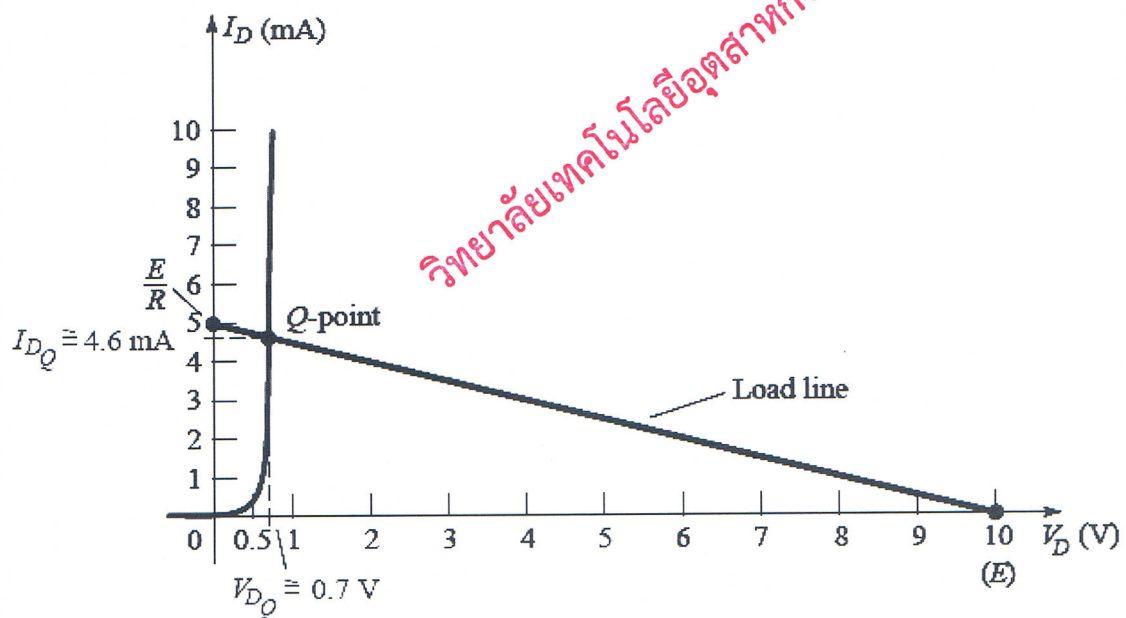
Part 2 (25 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

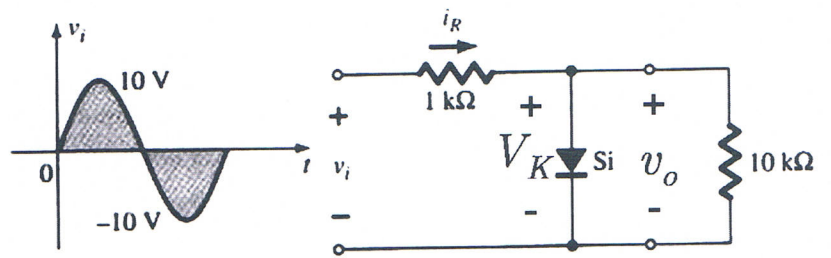
1. Sketch the output waveform of the network below.



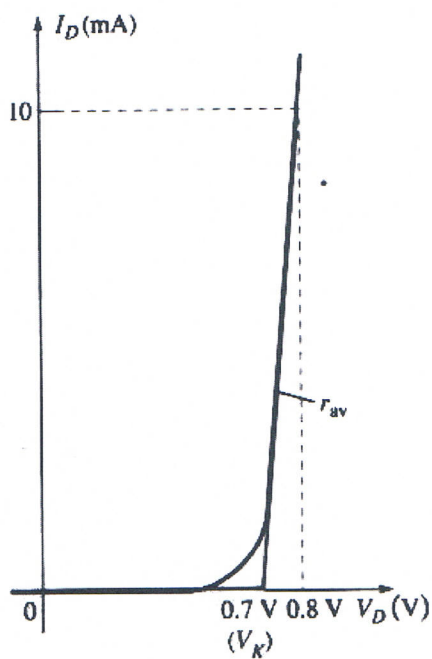
2. From the load line and diode characteristic curve, determine the value of R , V_R and E in the circuit below.



3. From figure below, sketch one cycle of output V_o and find the current I_R .

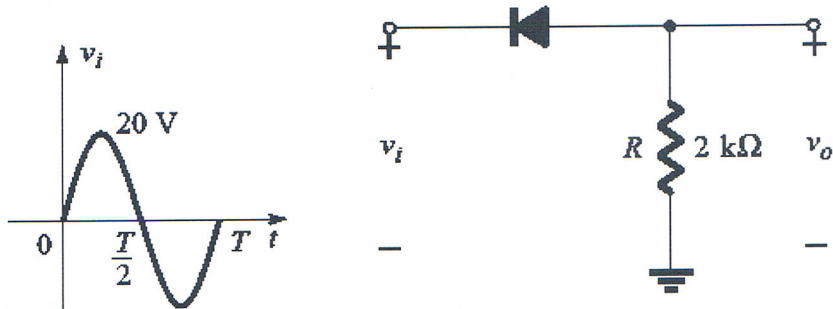


4. From the characteristic curve of diode below, draw the equivalent circuit of the diode.



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5. From the circuit, sketch the output and determine dc output voltage.



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Answer Sheet

เลขที่นั่งสอบ

Name: _____

ID: _____

Subject: 353152 Electronic Circuits I

	a	b	c	d
1.				
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